# Prajwal B N

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## EDUCATION

**Rajarajeswari College of Engineering** Bengaluru, IN
Bachelor of Engineering in Information Science and Engineering Nov 2022 - Present
CGPA: 8.35

**BGS PU College** Gauribidanur, IN
Science PCMC May 2020 - Mar 2022
Percentage: 81.66

**Sree Ramakrishna Sharadadevi Vidyamandira School** Gauribidanur, IN
10th ICSE Mar 2020
Percentage: 74.2

## EXPERIENCE

**Hindustan Aeronautics Limited (HAL)(Internship)** | ***Frontend*:** React, JavaScript, *Tailwind CSS, Axios* Bengaluru, IN
***Backend:*** *Python, Flask, SQLite, Flask-JWT-Extended, OpenCV, PyTorch, YOLOv8* Sep 2025 – Nov 2025

- Designed an AI-based Drone Surveillance System with Real-Time Object Detection using YOLOv8 and OpenCV.
- Implemented end-to-end full-stack modules such as real-time video streaming, geo-fencing, intrusion alerts, and AI analytics dashboard.
- Implemented PDF and CSV reporting, heatmap visualization, object detection logs, and JWT-based authentication.
- Optimized the Flask-based backend for low-resource environments and provided production-ready deployment documents.

## PROJECTS

**Sanjeevani - Medical Healthcare** | ***Frontend****: React.js, CSS3, Chart.js, Three.js, Web Speech API* May 2025
***Backend****: Python, FastAPI, JWT, Geolocation services, Twilio, JSON*

- Developed a secure healthcare system that allowed hospitals and patients to share medical information without experiencing security breaches.
- Integrated GPS tracking of ambulances and Twilio alerts to reduce emergency response times by 40%.
- Processed 5,000+ medical records using the FastAPI framework to optimize data workflow.

**Agropro - AI - Powered Smart Agricultural Advisory Platform** | *Flask, SQLite, HTML/CSS/JS, PyTorch, Scikit-learn, ResNet architecture, Weather APIs, SQLite, Pandas, NumPy* June 2025

- An AI-based agriculture intelligence platform that offers personalized crop, fertilizer, yield, price, and disease recommendations through machine learning algorithms and satellite-based analytics.
- Architected a full-stack agriculture intelligence platform using Flask and ML algorithms such as Random Forest, XGBoost-like algorithms, and ResNet-based CNN using real-time weather API and multilingual support (English, Kannada, Hindi).
- Designed smart farm management features such as irrigation planning, pest management recommendations, expense tracking, farm calendar, price monitoring, government scheme notifications, and a forum for farmers with real-time notifications.

## CERTIFICATIONS

**Web Development Fundamentals** | IBM SkillsBuild Mar 2025
**Geodata Processing using Python and Machine Learning** | IIRS Outreach Programme, ISRO April 2025
**SENTINEL HACK 5.0** | K.S. Institute of Technology April 2025

## SKILLS

**Programming Languages:** Java, Python, JavaScript (ES6+), C++, SQL, HTML5, CSS3
**Frameworks & Backend:** Flask, FastAPI, Django, Node.js, Express.js
**Frontend & Visualization:** React.js, Bootstrap, Chart.js, Three.js
**Databases:** MySQL, MongoDB
**Cloud & DevOps:** Render, Git, GitHub
**Machine Learning & AI:** PyTorch, Scikit-learn, Pandas, NumPy, SciPy, JAX
**Data Analysis & Visualization:** Matplotlib, Seaborn, Data Cleaning, Statistical Modeling

## Achievements

Secured 1st prize at Department Project Expo and KSCST funding (₹4,000) for Agropro - AI - Powered Smart Agricultural Advisory Platform (2025-2026)